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Lab 9.3 Machine Learning 2 - Machine Learning Implementation to Edge Device

Building Up The Entire Monitoring System

3.1 MTConnect Data Stream

Let's move on to the last step for the entire smart IIoT monitoring system. In this section, we will integrate each monitoring part we practiced in all previous labs into a single system to realize a real-time web-based monitoring system as Figure 1. In the previous section of this lab, we practiced the use of the machine learning model using TensorFlow on Raspberry Pi. In this part, let's make MTConnect data stream including ML results based on the model. The schematic and the data items are shown in Figure 7.

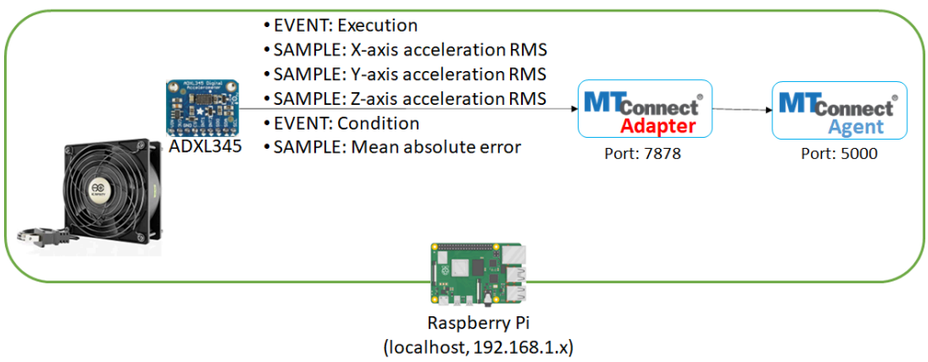

Figure 7 Schematic and data items of MTConnect

The examples of 'agent.cfg' and 'device.xml' are Figure 8 and Figure 9, respectively. You may have different a XML data structure or data item definitions. Also, you can check the example of MTConnect agent running on the server computer: http://mepotrb16.ecn.purdue.edu:5000/.

```
Devices = device.xml
Port = 5000
AllowPut = true
ReconnectInterval = 1000
BufferSize = 17
MonitorConfigFiles = true
Pretty = true
SchemaVersion = 1.5

Adapters {
   # Log file has all machines with device name prefixed
   ADXL345 {
      Host = 127.0.0.1
      Port = 7878
   }
}
```

Figure 8 front part of 'agent.cfg'

Task 3.1

Run MTConnect agent on Raspberry Pi with the XML structure and data items defined above.

- Please note that the command to run MTConnect agent is `sudo ./agent` in the same directory of the execution file.


```xml
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectDevices xmlns:m="urn:mtconnect.org:MTConnectDevices:1.1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns=
"urn:mtconnect.org:MTConnectDevices:1.1" xsi:schemaLocation="urn:mtconnect.org:MTConnectDevices:1.1 http://www.mtconnect.org/schemas/MTConnectDevices_1.1.xsd">
    <Header creationTime="2010-03-04T18:44:40+00:00" sender="localhost" instanceId="1267728234" bufferSize="131072" version="1.1"/>
    <Devices>
        <Device id="adxl345" name="ADXL345" uuid="001">
            <DataItems>
                <DataItem category="EVENT" id="001" name="avail" type="AVAILABILITY"/>
                <DataItem category="EVENT" id="e1" name="execution" type="EXECUTION"/>
                <DataItem category="EVENT" id="c1" name="condition" type="STATE"/>
                <DataItem category="SAMPLE" id="error1" name="mae" type="LOSS"/>
                <DataItem category="SAMPLE" id="a1" name="Xacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
                <DataItem category="SAMPLE" id="a2" name="Yacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
                <DataItem category="SAMPLE" id="a3" name="Zacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
            </DataItems>
        </Device>
    </Devices>
</MTConnectDevices>
```


Figure 9 'device.xml'

- The '*agent.cfg*' and '*device.xml*' must be in the same directory.
- To apply the schema and style in MTConnect agent, '*/schema*' and '*/styles*' must be in the same directory as well.

Capture '/current' response from the MTConnect agent as Figure 10 and attach it to the report.

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index.i | L9_Col | L9_Col | Home | M | Ask Gemini

Not secure 192.168.1.23:5000/current

- creationTime: 2026-04-01T20:40:19Z
- sender: AaryaFarheen
- instanceId: 1775075927
- version: 1.8.0.3
- bufferSize: 131072
- nextSequence: 10
- firstSequence: 1
- lastSequence: 9

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-01T20:38:47.593513Z	Acceleration		Xacc	a1	3	UNAVAILABLE
2026-04-01T20:38:47.593957Z	Acceleration		Yacc	a2	7	UNAVAILABLE
2026-04-01T20:38:47.593789Z	Acceleration		Zacc	a3	5	UNAVAILABLE
2026-04-01T20:38:47.593666Z	Loss		mae	error1	4	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-01T20:38:47.594124Z	Availability		avail	001	9	UNAVAILABLE
2026-04-01T20:38:47.593867Z	State		condition	c1	6	UNAVAILABLE
2026-04-01T20:38:47.594048Z	Execution		execution	e1	8	UNAVAILABLE
2026-04-01T20:38:47.593438Z	AssetChanged			s1_asset_chg	2	UNAVAILABLE
2026-04-01T20:38:47.592474Z	AssetRemoved			s1_asset_rem	1	UNAVAILABLE

- creationTime: 2022-04-01T03:31:33Z
- sender: lamm-UltraPoint
- instanceId: 1648783879
- version: 1.8.0.2
- bufferSize: 131072
- nextSequence: 10
- firstSequence: 1
- lastSequence: 9

Device: ADXL345; UUID: 001

Device : ADXL345

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-04-01T03:31:19.501339Z	Acceleration		Xacc	a1	2	UNAVAILABLE
2022-04-01T03:31:19.502042Z	Acceleration		Yacc	a2	7	UNAVAILABLE
2022-04-01T03:31:19.501693Z	Acceleration		Zacc	a3	4	UNAVAILABLE
2022-04-01T03:31:19.501559Z	Loss		mae	error1	3	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-04-01T03:31:19.502272Z	Availability		avail	001	9	UNAVAILABLE
2022-04-01T03:31:19.500637Z	AssetChanged			adxl345_asset_chg	1	UNAVAILABLE
2022-04-01T03:31:19.501909Z	AssetRemoved			adxl345_asset_rem	6	UNAVAILABLE
2022-04-01T03:31:19.501816Z	State		condition	c1	5	UNAVAILABLE
2022-04-01T03:31:19.502177Z	Execution		execution	e1	8	UNAVAILABLE

Figure 10 MTConnect agent '/current' response without MTConnect adapter

The next is to run MTConnect adapter. The sample MTConnect adapter based on the previous section is prepared, lab9_adapter_sample.py. The sample is incomplete as is. **Please note that to run MTConnect adapter in this case, 'data_item.py' and 'mtconnect_adapter.py' modules as the previous labs. In addition, the '/models' and sub-directory for your autoencoder model must be in the same directory of the MTConnect adapter Python script**. Perform Task 3.2 to finish MTConnect data stream.

Task 3.2

Complete the sample script (lab9_adapter_sample.py) and then run MTConnect adapter.

- You need to modify the variables wrapped in asteriks below. Also, you need to change the '???' parameters to the appropriate parameters.
- If you have a different XML structure and data items from the example, you must change other parts of the sample script as well.

Python - Python3 ("Main of lab9_adapter_sample.py)

```
if __name__ == "__main__":
    *model_path* = "models/YourModelDirectory/" # model file directory, you must change this
    *model* = tf.keras.models.load_model(model_path) # load the model from the path above
    *threshold* =  # float: threshold (MAE loss) for the ML model
    *min_val* =  # float: minimum value for normalization
    *max_val* =  # float: maximum value for normalization

    # start MTConnect Adapter
    MTConnectAdapter(???, ???) # Args: host ip, port number
```

Python - Python3 (Part of "MTConnectAdapter" class of lab9_adapter_sample.py)

```
    def adapter_stream(self):
        while True:
            try:
                x, y, z = measureData(acc,1000) # x=x-axis, y=y-axis, z-axis acceleration ar
                *input_feature* = # your input feature
                input_feature_normalized = tensorNormalization(input_feature, ???, ???) # no
```

```
result = predict(model, *FINAL_FEATURE_INPUT*, threshold)
x_rms = timeFeatures(x)[2] # calculate rms of x-axis
*y_rms* =  # calculate rms of y-axis
*z_rms* =  # calculate rms of z-axis
mae = result[?] # mean absolute error (loss) of the model

if *AFF is running*:
    execution = 'ACTIVE'
    if *AFF is normal*:
        condition = 'NORMAL'
    else:
        condition = 'ABNORMAL'
else:
    execution = 'STOPPED'
    condition = 'UNAVAILABLE'

now = datetime.datetime.now() # get current data time

self.adapter.begin_gather() # start to collection

self.Xacc.set_value(str(???))
self.Yacc.set_value(str(???))
self.Zacc.set_value(str(???))
self.execution.set_value(str(???))
self.condition.set_value(str(???))
self.mae.set_value(str(???))

self.adapter.complete_gather() # end of collection
```

Task 3.3

Capture ‘/current’ response from the MTConnect agent as Figure 11 and attach it to the report.

- Confirm if every data item works as expected by running the AFF.

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index.i | L9_Col | L9_Col | Home | M | Ask Gemini

Not secure 192.168.1.23:5000/current

- creationTime: 2026-04-01T21:08:50Z
- sender: AaryaFarheen
- instanceId: 1775077512
- version: 1.8.0.3
- bufferSize: 131072
- nextSequence: 37
- firstSequence: 1
- lastSequence: 36

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-04-01T21:08:49.85488Z	Acceleration		Xacc	a1	33	1.81253210967681
2026-04-01T21:08:49.85488Z	Acceleration		Yacc	a2	34	1.98741947306839
2026-04-01T21:08:49.85488Z	Acceleration		Zacc	a3	35	10.2785749729179
2026-04-01T21:08:49.85488Z	Loss		mae	error1	36	0.09823594

Events

- creationTime: 2022-04-01T03:51:43Z
- sender: lamm-UltraPoint
- instanceId: 1648783879
- version: 1.8.0.2
- bufferSize: 131072
- nextSequence: 52
- firstSequence: 1
- lastSequence: 51

	Sequence	Value
	10	AVAILABLE
	16	ABNORMAL
	15	ACTIVE
chg	2	UNAVAILABLE
rem	1	UNAVAILABLE

Device: ADXL345; UUID: 001

Device : ADXL345

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-04-01T03:51:43.15467Z	Acceleration		Xacc	a1	48	0.822723114117054
2022-04-01T03:51:43.15467Z	Acceleration		Yacc	a2	49	0.858075038336765
2022-04-01T03:51:43.15467Z	Acceleration		Zacc	a3	50	9.67068542083701
2022-04-01T03:51:43.15467Z	Loss		mae	error1	51	0.032767966

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-04-01T03:51:31.048541Z	Availability		avail	001	10	AVAILABLE
2022-04-01T03:31:19.500637Z	AssetChanged			adxl345_asset_chg	1	UNAVAILABLE
2022-04-01T03:31:19.501909Z	AssetRemoved			adxl345_asset_rem	6	UNAVAILABLE
2022-04-01T03:31:19.501816Z	State		condition	c1	5	UNAVAILABLE
2022-04-01T03:51:32.217381Z	Execution		execution	e1	14	STOPPED

Figure 11 MTConnect agent '/current' response with MTConnect adapter

Please continue to [Lab 9.4 here](#).